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Proof that $\delta(\omega \to \omega_0) = \pi/2$

Let $\omega = \omega_0 \mp \epsilon$. $\Rightarrow \quad \omega_0 - \omega = \pm\epsilon$

$$\omega_0^2 - \omega^2 = (\omega_0 + \omega)(\omega_0 - \omega) = \underbrace{(2\omega_0 \mp \epsilon)}_{\approx 2\omega_0 \text{ for small } \epsilon.}(\pm\epsilon)$$

$$= \boxed{\pm 2\omega_0 \epsilon}$$

$$\frac{\gamma\omega}{\omega_0^2 - \omega^2} = \frac{\gamma(\omega_0 \mp \epsilon)}{\omega_0^2 - (\omega_0 \mp \epsilon)^2} \approx \frac{\gamma\omega_0}{\pm 2\omega_0 \epsilon} = \boxed{\pm \frac{\gamma}{2\epsilon}}$$

$$\Rightarrow \boxed{\tan\delta \approx \pm \frac{\gamma}{2\epsilon}} \Rightarrow \cot\delta = \frac{1}{\tan\delta} \approx \pm\frac{2\epsilon}{\gamma}$$

But, $\cot\delta = \dfrac{\cos\delta}{\sin\delta} = \dfrac{\sin\left(\frac{\pi}{2} - \delta\right)}{\cos\left(\frac{\pi}{2} - \delta\right)} = \tan\left(\frac{\pi}{2} - \delta\right)$

[$\because \tan\delta \approx \delta$ for $|\delta| \ll 1$]

$$\Rightarrow \tan\left(\frac{\pi}{2} - \delta\right) \approx \pm\frac{2\epsilon}{\gamma} \Rightarrow \frac{\pi}{2} - \delta \approx \mp\frac{2\epsilon}{\gamma} \Rightarrow$$

$$\boxed{\delta = \frac{\pi}{2} \mp \frac{2\epsilon}{\gamma} \xrightarrow[\epsilon \to 0]{} \frac{\pi}{2}}$$

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Small freq. limit : $\omega \ll \omega_0$

$$A(\omega) = \frac{f_0}{\left[(\omega^2-\omega_0^2)^2 + \gamma^2\omega^2\right]^{1/2}} = \frac{f_0}{\sqrt{D(\omega)}}$$

where,

$$D(\omega) = (\omega^2-\omega_0^2)^2 + \gamma^2\omega^2$$

It can be shown that

$$A(\omega) \simeq \underbrace{\frac{f_0}{\omega_0^2}}_{A(0)}\left[1 + c\omega^2\right] = A(0)(1+c\omega^2),$$

where $c \simeq \frac{\omega_m^2}{\omega_0^4} = \frac{1}{\omega_0^2}\left(1 - \frac{\gamma^2}{2\omega_0^2}\right)$

For small damping, i.e, for $\gamma \ll \omega_0$

$c \simeq 1/\omega_0^2$ or

$$A(\omega) \simeq A(0)\left[1 + \omega^2/\omega_0^2\right], \quad \text{for } \omega \ll \omega_0$$

$\therefore$ Near $\omega=0$, $A(\omega)$ parabolic.

Large freq. limit : $\omega \gg \omega_0$

Assumed : $\gamma \ll \omega_0$

$$\Rightarrow \omega^2 - \omega_0^2 \simeq \omega^2 \Rightarrow (\omega^2-\omega_0^2)^2 \simeq \omega^4$$

$$\Rightarrow D(\omega) = (\omega^2-\omega_0^2)^2 + \gamma^2\omega^2$$

$$\simeq \omega^4 + \gamma^2\omega^2$$

$$= \omega^4(1 + \gamma^2/\omega^2)$$

$$\simeq \omega^4 \Rightarrow \sqrt{D(\omega)} \simeq \omega^2$$

$$A(\omega) \xrightarrow[\omega\to\infty]{} \sim \frac{f_0}{\omega^2}$$

As $\omega \to \infty$, $A(\omega) \sim \frac{f_0}{\omega^2} \to 0$.

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Proof that $A(\omega)$ is maximum at $\omega_m = \sqrt{\omega_0^2 - \gamma^2/2}$

Let $D(\omega) = (\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2$, $\quad A(\omega) = \dfrac{f_0}{\sqrt{D(\omega)}}$

$$A'(\omega) = -\frac{f_0}{2[D(\omega)]^{3/2}} D'(\omega) = 0 \Rightarrow D'(\omega) = 0$$

$$D'(\omega) = 2(\omega^2 - \omega_0^2)(2\omega) + 2\gamma^2\omega = 0$$

$$\Rightarrow 2\omega\left[2(\omega^2 - \omega_0^2) + \gamma^2\right] = 0 \Rightarrow$$

Non-trivial Root: $\omega^2 = \omega_0^2 - \dfrac{\gamma^2}{2} \equiv \omega_m^2$

At $\omega = \omega_m$, $D(\omega_m) = (-\gamma^2/2)^2 + \gamma^2(\omega_0^2 - \gamma^2/2)$

$$= \gamma^2\underbrace{(\omega_0^2 - \gamma^2/4)}_{\omega_d^2} = \gamma^2\omega_d^2$$

$$A_m = A(\omega_m) = f_0/\sqrt{D(\omega_m)} = \frac{f_0}{\gamma\omega_d} = \frac{f_0}{\omega_0^2}\frac{\omega_0}{\gamma}\frac{\omega_0}{\omega_d} = \frac{f_0}{\omega_0^2}\frac{Q}{[1 - 1/4Q^2]^{1/2}}$$

$$\boxed{\frac{A_m}{A_0} = \frac{A(\omega_m)}{A(0)} = \frac{Q}{[1 - 1/4Q^2]^{1/2}}}$$

$$\omega_m^2 = \omega_0^2 - \frac{\gamma^2}{2}$$
$$= \omega_0^2\left[1 - \frac{1}{2}\frac{\gamma^2}{\omega_0^2}\right]$$
$$= \omega_0^2[1 - 1/2Q^2]$$

$$\omega_d^2 = \omega_0^2 - \gamma^2/4$$
$$= \omega_0^2(1 - \gamma^2/4\omega_0^2)$$
$$= \omega_0^2(1 - 1/4Q^2)$$

$$\frac{\omega_d}{\omega_0} = (1 - 1/4Q^2)^{1/2}$$

We can also see this by completing squares:

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$$D(\omega) = (\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2$$

$$= (\omega^2)^2 - 2\underbrace{\left[\omega_0^2 - \frac{\gamma^2}{2}\right]}_{\omega_m^2}\omega^2 + (\omega_0^2)^2$$

$$= (\omega^2 - \omega_m^2)^2 + (\omega_0^2)^2 - (\omega_m^2)^2$$

$$= (\omega^2 - \omega_m^2)^2 + \underbrace{\omega_d^2\gamma^2}_{\omega_0^2 - \gamma^2/4} \Rightarrow$$

$$(\omega_0^2 + \omega_m^2)(\omega_0^2 - \omega_m^2)$$
$$= \left(2\omega_0^2 - \frac{\gamma^2}{2}\right)\left(\frac{\gamma^2}{2}\right)$$
$$= \gamma^2\underbrace{(\omega_0^2 - \gamma^2/4)}_{\omega_d^2}$$

$\therefore$ Min. value of $D(\omega)$ occurs at $\omega = \omega_m$:

$$D(\omega_m) = \omega_d^2\gamma^2 \Rightarrow \sqrt{D(\omega_m)} = \omega_d\gamma$$

$$\Rightarrow A(\omega_m) = \frac{f_0}{\omega_d\gamma} \Rightarrow \frac{f_0}{\omega_0^2}\frac{\omega_0/\gamma}{[1 - \gamma^2/4\omega_0^2]^{1/2}} = A(0)\frac{Q}{[1 - 1/4Q^2]^{1/2}}$$